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TECHNICAL TEAM

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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROGRAMMING PROJECT PRESENTATION

Project Title: [Vehicle Service Reminder and Maintenance Tracker]

Course: java Programming Mini Project

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Department: Computer Science

Mentor / Guide: [Aravind]

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INTRODUCTION

The Vehicle Service Reminder and Maintenance Tracker is a Java-based application designed to help users manage vehicle maintenance efficiently. It provides timely service reminders, stores maintenance records, and allows easy access to vehicle history through a simple interface.

In today's busy life, people often forget service schedules, leading to poor vehicle performance and higher costs. This project offers a practical solution by ensuring timely maintenance and better organization of service records.

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OBJECTIVE

The main objective of this project is to design and implement an efficient program to automate vehicle service tracking and maintenance activities. It aims to reduce the difficulty of manually remembering service schedules by providing a systematic approach to managing vehicle-related information.

The project focuses on applying core programming concepts such as logical thinking, modular design, file handling, and date processing. It also emphasizes the use of a user-friendly interface to ensure that the system is easy to understand and operate for all users.

Another important objective is to demonstrate problem-solving and analytical skills by developing a system that can accurately calculate service dates, generate reminders, and maintain service records. Overall, the project aims to provide a reliable and practical solution for improving vehicle maintenance management.

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PROBLEM STATEMENT

In the current scenario, many vehicle owners rely on manual methods to track service schedules, which often leads to missed maintenance and poor record management. There is a lack of simple and accessible systems that can provide timely reminders and organize vehicle service data efficiently.

Existing methods have several limitations, such as difficulty in maintaining records, absence of automatic reminders, and lack of easy search or retrieval of past data. These inefficiencies can result in delayed servicing, increased maintenance costs, and reduced vehicle performance.

To address these challenges, this project aims to develop a user-friendly application that automates vehicle service tracking. It provides timely reminders, stores service records systematically, and allows easy viewing and searching of data, thereby improving overall vehicle maintenance management.

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SYSTEM REQUIREMENTS

Hardware Requirements:

Processor: Intel i3 or higher

RAM: 4 GB minimum

Storage: 500 MB free space

Software Requirements:

Operating System: Windows / Linux

Programming Language: Java

IDE: Eclipse IDE

JDK: Java Development Kit (JDK 8 or higher)

Libraries Used: Java Swing, Java Time API

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SYSTEM DESIGN

- **Algorithm outline**

Start

Enter vehicle details

Convert input to date format

Calculate next service date

Get current date

Compare dates

Display reminder message

Save or view/search records if required

End

- **Flowchart (logic flow)**

Start →Input Vehicle Details →Process Data → Calculate Next Service Date →Compare with Current Date →Display Reminder (Due / Today / Overdue) →Save/View/Search Option →End

- **Block diagram (system modules)**

Input Module →Processing Module →Reminder Module →Storage Module →Display Module → Search Module

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MODULE DESCRIPTION

Input Module – Takes user input such as vehicle name, service date, and interval

Processing Module – Calculates next service date and checks service status

Reminder Module – Displays reminder messages (Due / Today / Overdue)

Output Module – Shows results to the user through GUI

File Handling Module – Saves and retrieves service records from file

Display Module – Displays records in table format

Search Module – Searches and filters records based on vehicle name

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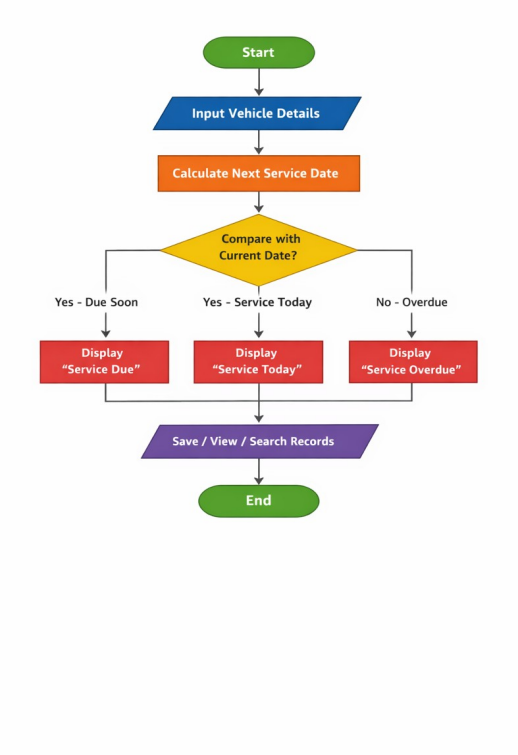
ALGORITHM / FLOWCHART

1. Start
2. Initialize variables
3. Input data from the user
4. Process data
5. Display output
6. Stop

(Insert flowchart image here)



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IMPLEMENTATION

Techniques used:

Loops – Used to read and display multiple records from the file

Conditional Statements – Used to check service status (Due / Today / Overdue)

Functions (Methods) – Used to organize code into separate tasks like display and search

Arrays – Used to store and process record data (split from file)

File Handling – Used to save and retrieve vehicle service records

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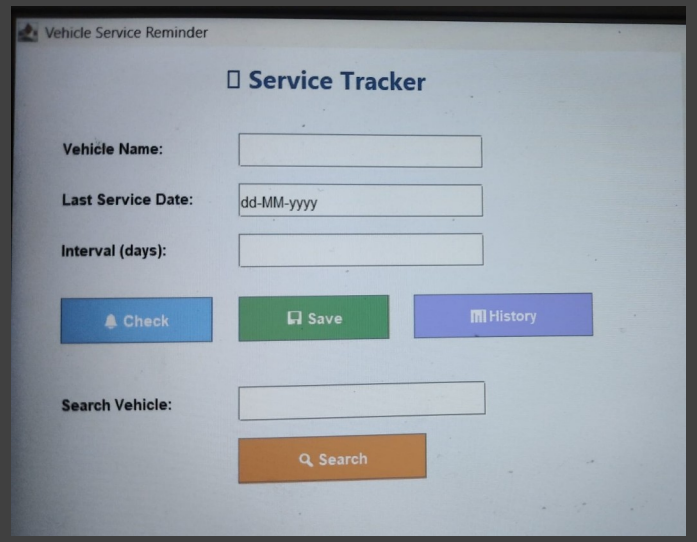
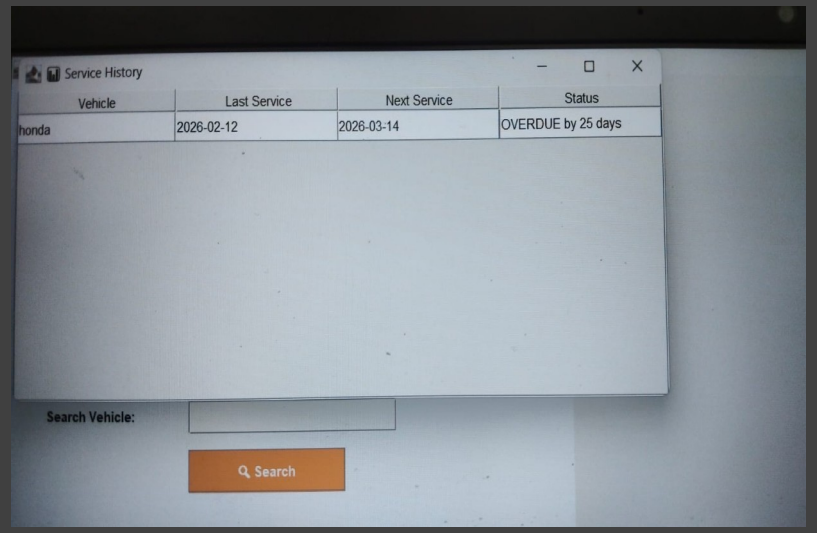
OUTPUT SCREENS

- Input screen



- Output history screen



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RESULTS & DISCUSSION

Result

The project was successfully implemented and tested. The system correctly calculates service dates, provides accurate reminders, and allows users to save, view, and search vehicle records efficiently.

Discussion

The project demonstrates how programming concepts can be used to solve real-life problems. It improves vehicle maintenance management through reminders and record handling. However, it has limitations like basic file storage and lack of real-time notifications, which can be improved in future.

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CONCLUSION & FUTURE ENHANCEMENT

Conclusion:

The project successfully achieved its objectives and improved understanding of programming concepts and system design.

Future Enhancements:

Add advanced GUI features

Integrate with a database

Extend functionality for better scalability

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REFERENCES / ACKNOWLEDGEMENT

References:

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Oracle – Java Documentation (Java Swing & Java Time API)

GeeksforGeeks – Tutorials on Java, Swing, and File Handling

W3Schools – Java Basics and Examples

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THANK YOU